Net change

1

a 40 kangaroos

b 8 months

2

a

i No **ii** Yes **iii** No **iv** Yes

b 3000

c

i Yes **ii** Yes **iii** No **iv** No

d 36 000

3

a $-2.02^\circ\text{C/minute}$ **b** -41.74°C **c** $-4.64^\circ\text{C/minute}$

4

a $-\dfrac{24}{\pi}$

b $-\dfrac{30}{\pi}$

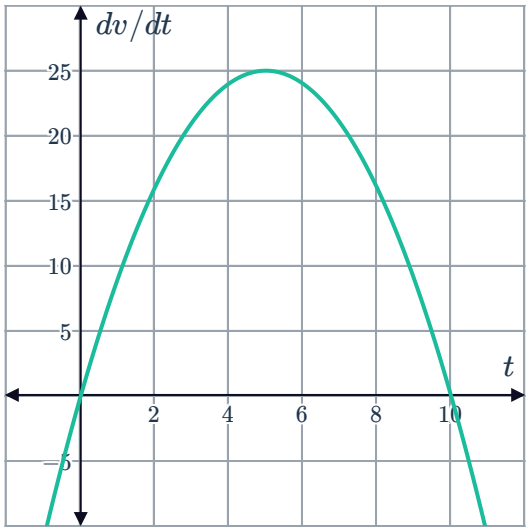
5

a 5818.82 L

b 175.71 L

6

a



b 25 L/min

c $V = 5t^2 - \dfrac{t^3}{3}$

d 15 minutes

7

a 3700 tonnes/year **b** 136 tonnes/year **c** 100 tonnes/year **d** 4140 tonnes

8

a $M = \dfrac{81t^2}{2} - \dfrac{t^4}{4} + 1000$

b 2568 kg

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c $t = 9$

9

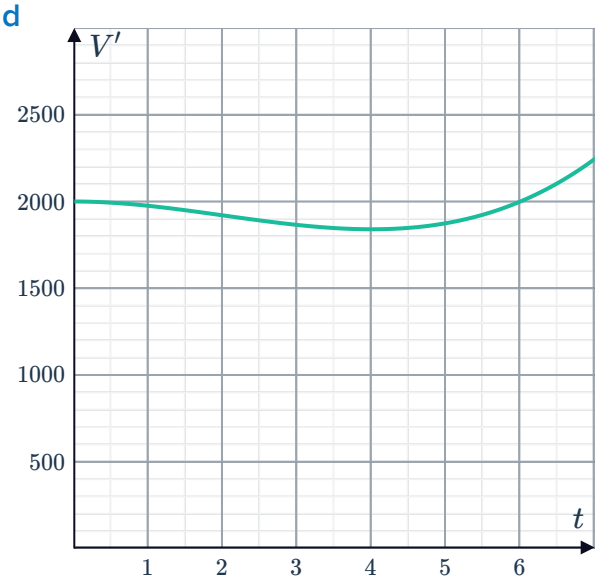
a 2000 L/h

b

t	-1	0	1	4	6
V''	75	0	-45	0	180

c $t = 7$ at 2245 L/h

d



e 5831.25 units^2

f The total change in the amount of water in the tank in the first 3 hours is 5831.25 L.

10

a $\dfrac{dV}{dt} = -25$

b $V = 15\,625 - 25t$

c $t = 625$

11

a $m = \dfrac{2t^{\frac{3}{2}}}{360} + \dfrac{t^3}{4320}$

b $\dfrac{176}{135} \text{ g}$

c 0.004 g

12

a 145 customers **b** 184 customers **c** 134 customers

Cost, revenue and profit applications

13

a $\$6222$

b $\$1037/\text{item}$

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14

a $P(x) = 0.0004x^3 + 0.6x^2 - 4.1x - 60$ **b** $\$2730.20$

c $\$9493.40$

15

a

i $40\,000 \text{ units}^2$ **ii** $32\,500 \text{ units}^2$

b $\$40\,000$

c $\$34\,500$ **d** $\$5500$

e $-\$5000$

16

a $C = 0.1x^4 + 2x^2 + 2000$ **b** $\$632\,000$

17

a $C(x) = 3x^3 - 35x^2 + 500x + \text{initial costs}$

b $\$532\,000$

18

a $C(x) = 240x^{\frac{1}{2}} + \text{initial costs}$ **b** $\$960$

19

a $-\$17\,601.30$ **b** $-\$704\,700$

c $-\$23\,490/\text{item}$

20

a $P(x) = 3x^2 + 2x^3 + 5x$ **b** $\$56\,850$

21

a $\$5.03$ **b** $\$4.57$

22

a $R = 401t - \dfrac{250}{(t+1)^2} + 250$ **b** $\$449\,611.11/\text{week}$

c $\$402\,842.40$

23

a

i $\$81.62$ **ii** $\$9000$ **iii** $\$90$

b

i $\$31.42$ **ii** $\$1607.70$ **iii** $\$16.08$

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